

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** of experience designing and shipping distributed systems at Google, YouTube, Microsoft, and Whatnot. Built infrastructure powering **YouTube** playback, **Google Search** ranking, and **Google Shopping** inventory serving billions of daily operations. Led end-to-end ownership of **C++** kernel networking at Microsoft and **Go/Python** microservices at Whatnot. Delivered **Kubernetes** migration reducing operational overhead by 40%. Expert in **distributed systems**, **Linux** performance, and low-latency system design. Partners with product and data teams to ship high-impact features at scale. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: C++, Python, Go, SQL, TypeScript, JavaScript, HTML, CSS

Infrastructure: Kubernetes, Docker, Linux, distributed systems

Data Stores: Bigtable, Spanner, MySQL, Cassandra

Networking: TCP/IP, kernel level networking, gRPC

Leadership: on call, technical roadmaps, cross-functional partnerships, Bash, Ansible, Terraform, Prometheus, Grafana, Jenkins, GitHub Actions

Work Experience

Education

B.S. Honors degrees in Computer Science

Virginia Tech | 2004 - 2007

Staff Software Engineer | YouTube

02/2025 - Present | Remote

- Own playback infrastructure serving 2B+ daily video streams, driving reliability improvements and latency reductions across **C++** and **Go** services.
- Architect **distributed systems** for video transcoding pipeline handling 500K+ concurrent jobs with **Kubernetes**-based autoscaling.
- Lead on-call rotation covering critical path services; reduced P1 incidents by 35% through improved alerting and runbook automation.
- Partner with video infrastructure and client teams to define performance budgets and ship low-latency playback experiences.
- Design data pipelines processing 10PB+ daily telemetry for quality monitoring using **Python** and **Spanner**.
- Drive technical hiring: define role requirements, conduct system design interviews, and mentor junior engineers on **Linux** internals.

Staff Software Engineer | Whatnot

05/2024 - 02/2025 | Remote

- Led infrastructure team building live marketplace platform processing 100K+ concurrent bidding sessions during peak drops.
- Delivered **Kubernetes** migration for 80+ microservices, reducing deployment cycle from 2 hours to 15 minutes.
- Architected real-time inventory system using **Go** and **Cassandra** supporting atomic transactions across 50K+ daily auctions.

- Built observability stack with Prometheus and Grafana; reduced MTTR from 45 minutes to 12 minutes.
- Designed multi-region failover strategy achieving 99.95% uptime during traffic spikes.
- Established on-call practices and incident response protocols; handled 20+sev1 incidents with 0 customer impact.

Staff Software Engineer/Manager | Google

03/2015 - 04/2024 | Remote

- Led team of 8 engineers owning **Google Search** ranking infrastructure serving 3.5B daily queries; drove ranking quality improvements.
- Designed next-generation serving system in **C++** cutting P99 latency by 30% while handling 10PB index across 50K+ machines.
- Built **Google Shopping** inventory pipeline processing 500M+ daily product updates with **Bigtable** and **Spanner**.
- Championed **Kubernetes** adoption across Search infrastructure; enabled horizontal pod autoscaling for 100+ services.
- Managed performance reviews and career development for 8 engineers; promoted 3 to senior and 2 to staff level.

Software Engineer | Google

02/2011 - 02/2015 | Remote

- Owned Search quality metrics pipeline processing 1TB+ daily logs; built alerting systems detecting ranking regressions within minutes.
- Developed **C++** components for query rewrite and intent classification serving 1B+ daily requests.
- Implemented distributed caching layer in **C++** reducing backend load by 25% during peak traffic.
- Contributed to open-source **Linux** performance tools used for kernel-level profiling of search serving infrastructure.
- Mentored 5 incoming engineers during onboarding; established code review practices for the team.

Software Design Engineer II | Microsoft

10/2007 - 12/2011 | Remote

- Designed and implemented core **TCP/IP** networking stack components in **C** for Windows kernel with **kernel level** optimizations.
- Developed Windows kernel filter drivers shipped in Windows 7 and Server 2008; maintained driver compatibility across service packs.
- Built automated testing framework for kernel network components catching 15+ critical bugs pre-release.
- Profiled and optimized **Linux**-equivalent networking paths; reduced context switch overhead by 18% in stress tests.

Software Design Engineer Intern | Microsoft

06/2006 - 08/2006 | Remote

- Built internal tooling in **C++** automating driver testing; reduced manual QA cycle from 3 days to 4 hours.
- Contributed bug fixes to Windows networking documentation and driver samples.
- Presented intern project to 50+ engineers demonstrating 40% improvement in test coverage.